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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

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1. Team & Project Overview

Team Name	404 Brain Not Found
Selected Domain	Domain 1 — The Aiversecity Future-Ready Workforce Domain
Selected SDG Track (1–17)	SDG 4 — Quality Education
Project / Solution Name	TrendPilot AI
Name (Leader)	Teng Ray Tzong
Email & Phone (Leader)	+60 11-31317873
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	2

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

The rapid adoption of artificial intelligence (AI) is changing job requirements and increasing the need for continuous reskilling. Workers may struggle to identify which skills to develop, while fresh graduates and career switchers may find it difficult to understand how their existing abilities relate to emerging career opportunities. Employers and university career centres also need accessible ways to support workforce development.

In Malaysia, employment statistics provide useful information about the overall labour market, but national vacancy figures alone cannot show whether a specific technical skill is becoming more or less relevant. Users therefore need career guidance that connects their current skills with possible career directions while clearly distinguishing verified labour-market evidence from AI-generated suggestions.

Without accessible guidance, individuals may spend time learning skills that do not align with their career interests or evolving job requirements. This can make career planning and lifelong learning more difficult.

TrendPilot AI addresses this challenge by helping users explore potential career paths, understand their skill gaps, and identify practical learning steps. It is designed as a career exploration and reskilling support tool, rather than a recruitment system or a tool that makes employment decisions.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

TrendPilot AI is a human-centred career exploration prototype that helps users connect their existing skills with potential future career paths.

Users enter their skills or career interests through a web interface. The Next.js, React and TypeScript frontend sends the input to a Python FastAPI backend through an HTTP JSON request. The backend validates the input and calls a large language model hosted through the Groq API using an OpenAI-compatible client.

The AI generates career-role suggestions, estimated match scores, matched skills, skills to develop, concise role descriptions, ordered learning pathways, and relevant United Nations Sustainable Development Goal (SDG) connections. The backend parses and normalises the response before returning structured results to the frontend.

The prototype also provides Malaysian employment indicators labelled with the Department of Statistics Malaysia (DOSM) as their source. These figures are currently a static published-data snapshot, not a live labour-market feed, and do not directly establish demand for individual technologies or skills.

The project supports SDG 4 (Quality Education) by encouraging skills development and lifelong learning. It also relates to SDG 8 through its focus on career exploration and workforce readiness.

The team intends to apply relevant skills from the SDC Phase 1 workshops, including GenAI API integration, structured request and response handling, and web application architecture. Further work will focus on integrating an updateable public data source, recording source and retrieval dates, evaluating recommendation usefulness, and presenting alternative future scenarios.

All AI-generated match scores are estimates based on limited user input. They are not verified employment probabilities, salary predictions, or hiring decisions. The user retains control over career choices.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend:

- Next.js, React and TypeScript for the web interface.
- Tailwind CSS for styling.
- HTTP/JSON requests to communicate with the backend.

Backend:

- Python and FastAPI for API endpoints, input validation and response handling.
- Uvicorn as the ASGI development server.
- Pydantic for request data validation.
- python-dotenv for loading environment variables.

AI Integration:

- Groq API through the OpenAI-compatible Python client.
- The default model configured in the current backend is openai/gpt-oss-20b; the actual model depends on the runtime configuration.
- GROQ_API_KEY is loaded from a local backend environment file and must not be committed to the repository.

Data:

- A static employment-statistics snapshot labelled with DOSM as the source.
- No database integration was identified in the submitted source code.

Deployment:

- The current project includes local development instructions for the frontend and backend.
- Public cloud deployment and a live prototype URL must be confirmed separately.

System flow: User → Next.js/React frontend → FastAPI backend → Groq-hosted LLM → JSON response processing → Career recommendations displayed in the frontend.

A separate /market-trends endpoint returns the currently hard-coded employment-statistics snapshot. No official SDC sponsor-provided API or cloud platform is claimed unless actual use can be verified.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

The team identified career planning and reskilling as the central user needs, focusing on working adults, fresh graduates, career switchers, and career-support organisations. The prototype translates these needs into a simple workflow: enter skills, explore possible career roles, review skill gaps, and follow an ordered learning pathway.

To keep the solution technically feasible, the team uses a modular web architecture with a Next.js frontend, a FastAPI backend, and an external LLM API. Structured JSON responses and backend validation help maintain a consistent interface between components.

Before Phase 2, the team plans to test representative sample profiles, verify the accuracy and usefulness of generated recommendations, improve error handling, and provide clear explanations of match-score limitations. The employment-data component will be strengthened with a verifiable, updateable public source and visible data dates. Alternative future scenarios will be labelled as scenarios rather than predictions unless supported by an appropriate forecasting method.

Deployment readiness will be confirmed through a working public prototype, secure API-key management, and accessible source-code documentation. User feedback will guide further interface and recommendation improvements.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG Target 4.4: By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	0 updateable external labour-market data feeds currently integrated in the prototype. Career recommendations are generated, but their usefulness has not yet been established through a documented user evaluation.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Percentage of test users who rate the career recommendations and learning pathways as useful; number of verified, dated labour-market data sources integrated.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Target at least 70% positive usefulness ratings in a small, documented user test, and integrate at least one verifiable, updateable public labour-market data source with source and retrieval dates.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [x] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [x] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [x] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [x] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next js	Code/Framework	MIT
React	Code/Framework	MIT
FastAPI	Code/Framework	MIT
Pydantic	Code / Library	MIT
OpenAI Python client	Code / Library	Apache-2.0
Groq-hosted openai/gpt-oss-20b	Model / API	Apache-2.0 for the model; API usage subject to provider terms
DOSM Employment Statistics, Q1 2026	Public data	Verify the applicable source terms before submission

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature] 

[Date] 24/10/2026